

CS & IT ENGINEERING



Computer Network - 1

Transport Layer

Lecture No. – 05

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Recap of Previous Lecture



Topic

TCP Connection Establishment





Topics to be Covered



Topic

TCP Data Transfer

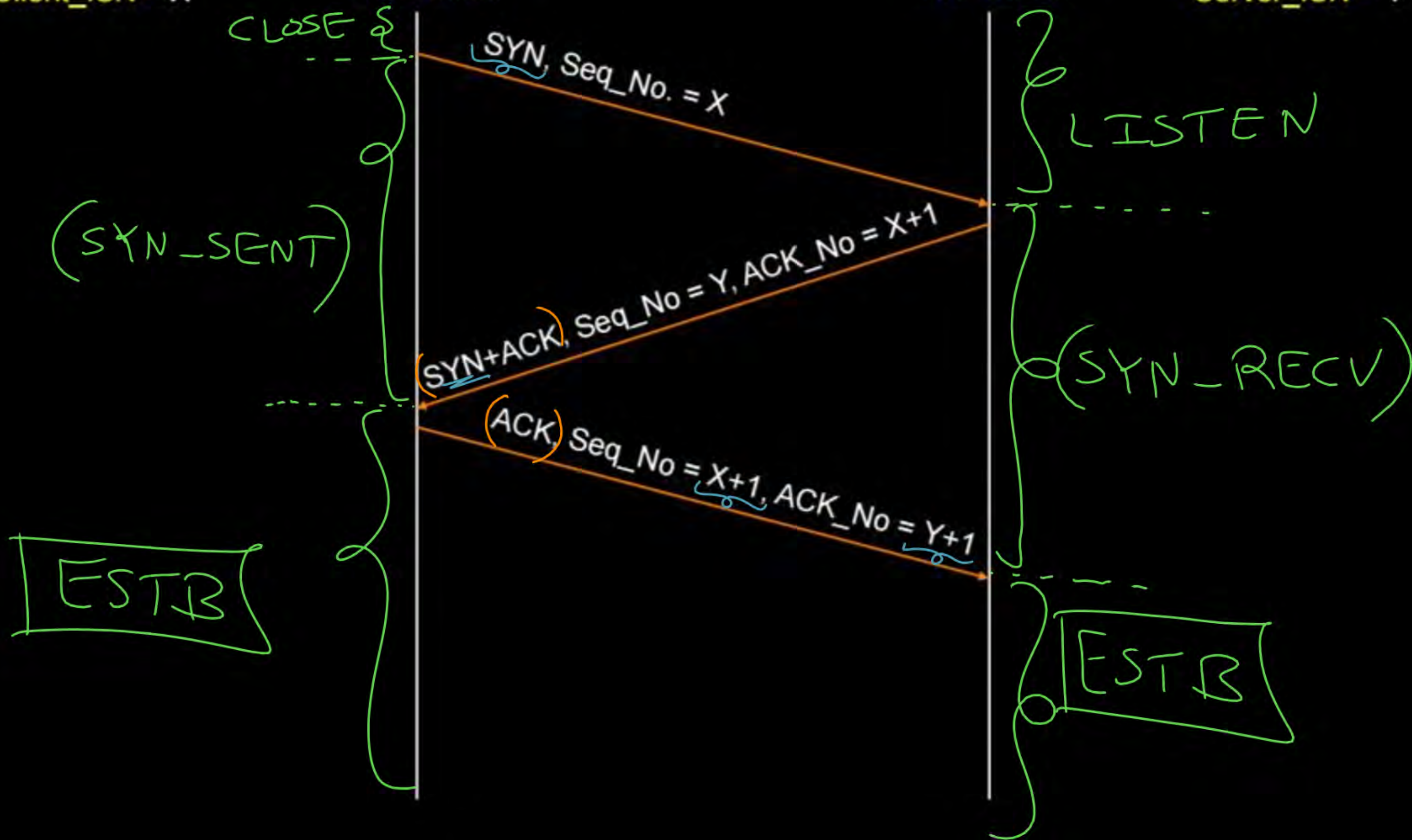


Client_ISN = X

TCP Client

TCP Server

Server_ISN = Y



[MCQ]

[GATE-IT-2008][1 Mark]



#Q. Which of the following statements are TRUE for 'three way handshake' for TCP connection establishment?

(S1) Loss of SYN + ACK from the server will not establish a connection TRUE

(S2) Loss of ACK from the client cannot establish the connection FALSE

(S3) The server moves LISTEN → SYN_RCVD → SYN_SENT → ESTABLISHED in the state machine on no packet loss FALSE

(S4) The server moves LISTEN → SYN_RCVD → ESTABLISHED in the state machine on no packet loss TRUE

A S2 and S3 only

C S1 and S3 only

✓ B S1 and S4 only

D S2 and S4 only

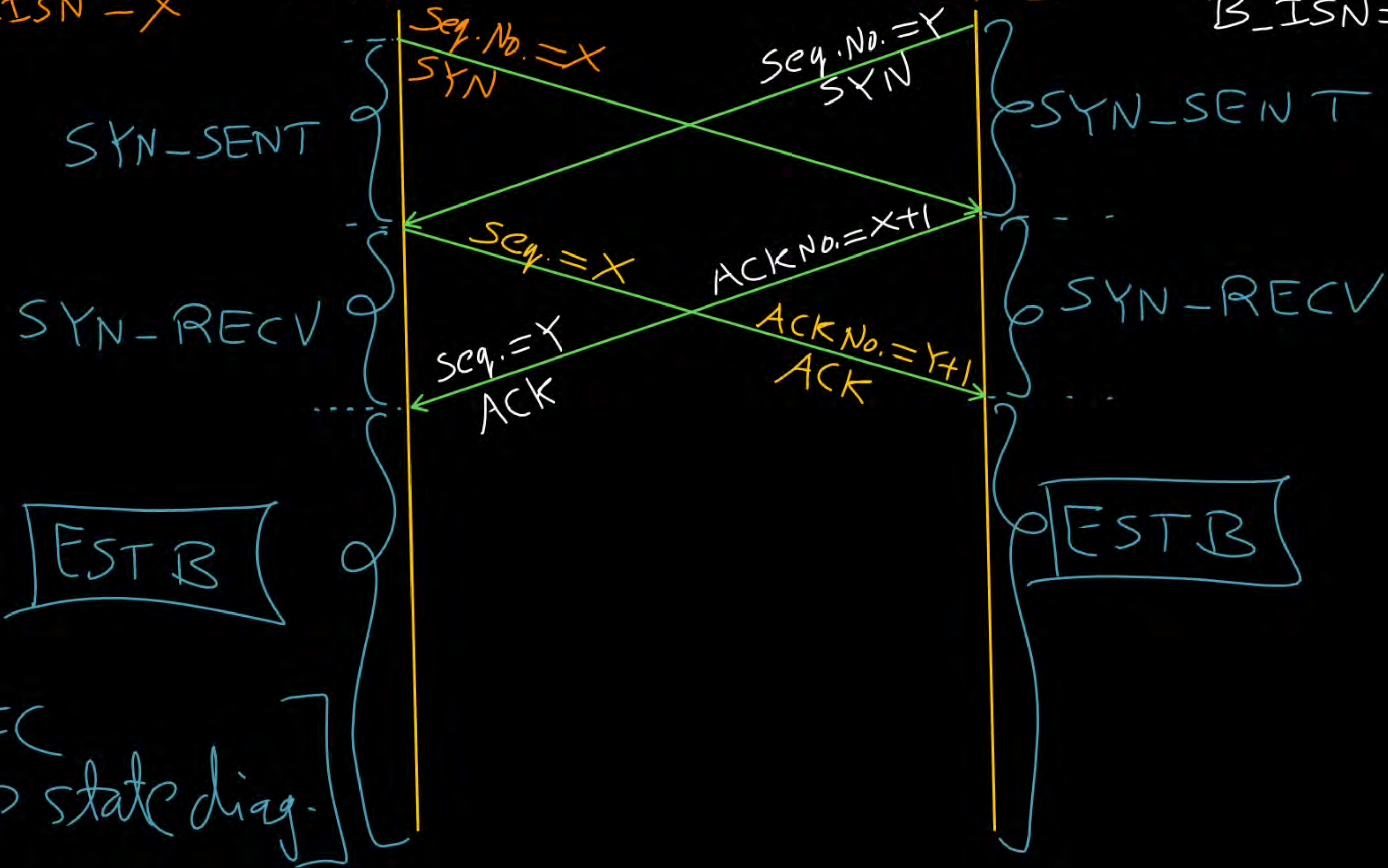
Ans: B

A_ISN = X

TCP A

TCP B

B_ISN = Y





Topic : TCP

TCP Data Transfer



TCP Client Process

P_1

SOCK_STREAM
(TCP Socket)

Send Buffer

Recv Buffer

TCP Server Process

P_2

SOCK_STREAM
(TCP Connection Socket 1)

Send Buffer

Recv Buffer

TCP

→ seq.no = Byte No.

* Conn. Estb.

* Conn. Estb.

TCP Client Process

P_1

SOCK_STREAM
(TCP Socket)

$$\begin{aligned}\text{LastByteWritten} &= P_1's \text{ ISN} = X \text{ (LBW)} \\ \text{LastByteSent} &= P_1's \text{ ISN} = X \text{ (LBS)} \\ \text{LastByteAcked} &= P_1's \text{ ISN} = X \text{ (LBA)}\end{aligned}$$

Send Buffer

(MaxSendBuffer = 1000 Bytes)



Recv Buffer

(MaxRecvBuffer = 1000 Bytes)



$$\begin{cases} \text{LastByteRead} &= P_2's \text{ ISN} = Y \\ \text{NextByteExpected} &= P_2's \text{ ISN} + 1 = (Y+1) \\ \text{LastByteRcvd} &= P_2's \text{ ISN} = Y \end{cases}$$

TCP Server Process

P_2

SOCK_STREAM
(TCP Connection
Socket 1)

Send Buffer

(MaxSendBuffer = 1200 Bytes)

Recv Buffer

(MaxRecvBuffer = 1200 Bytes)

LastByteWritten = P_2 's ISN = Y

LastByteSent = P_2 's ISN = Y

LastByteAcked = P_2 's ISN = Y

LastByteRead = P_1 's ISN = X

NextByteExpected = P_1 's ISN + 1 = X + 1

LastByteRcvd = P_1 's ISN = X



Topic : TCP

Message(Data) size = N byte



=> The process P_1 wants to write N bytes in 'Send Buffer'

if [(LastByteWritten - LastByteAked) + N] > MaxSendBuffer
→ Not allowed to write immediately, wait for sometime.

else

→ Allowed to write.

→ LastByteWritten = LastByteWritten + N;

Message₁ ⇒ 150 byte (N=150)
Message₂ ⇒ 200 byte (N=200)

$$ISN = x = 299$$

TCP Client Process

P_1

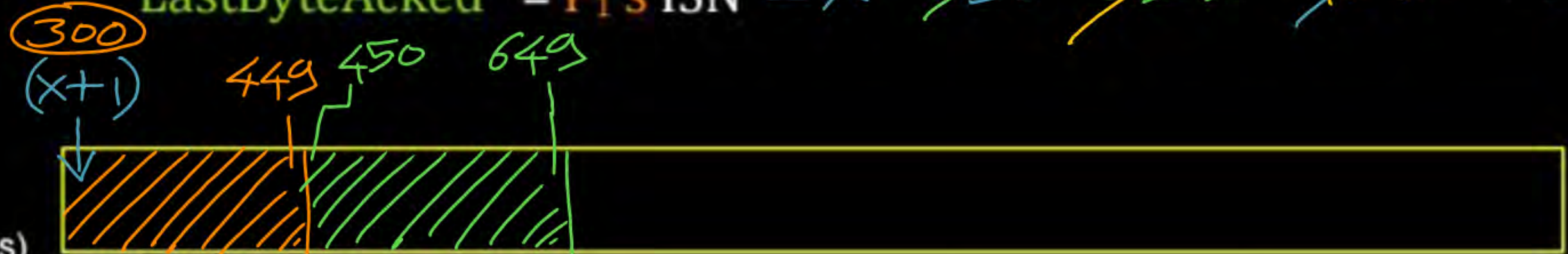
SOCK_STREAM
(TCP Socket)

Send Buffer

(MaxSendBuffer = 1000 Bytes)

Recv Buffer

(MaxRecvBuffer = 1000 Bytes)



$$\text{LastByteWritten} = P_1's \text{ ISN} = x = 299 = 449 = 649$$

$$\text{LastByteSent} = P_1's \text{ ISN} = x = 299 = 399 = 499 = 599$$

$$\text{LastByteAcked} = P_1's \text{ ISN} = x = 299 = 399 = 499 = 599$$



$$\text{LastByteRead} = P_2's \text{ ISN} = Y$$

$$\text{NextByteExpected} = P_2's \text{ ISN} + 1 = Y + 1$$

$$\text{LastByteRcvd} = P_2's \text{ ISN} = Y$$



Topic : TCP

Assume $MSS = 100 \text{ byte}$



=> Maximum Segment Size (MSS) : Maximum bytes can be kept in the payload field of a TCP segment

=> TCP protocol wants to transmit next segment from process P₁ 'Send Buffer'

→ Fill upto MSS bytes [(LastByteSent + 1) to (LastByteSent + MSS)]
in the payload field of segment from 'Send Buffer'

→ Sequence Number = LastByteSent + 1;

→ ACK Number = NextByteExpected;

→ LastByteSent = LastByteSent + MSS; ✓



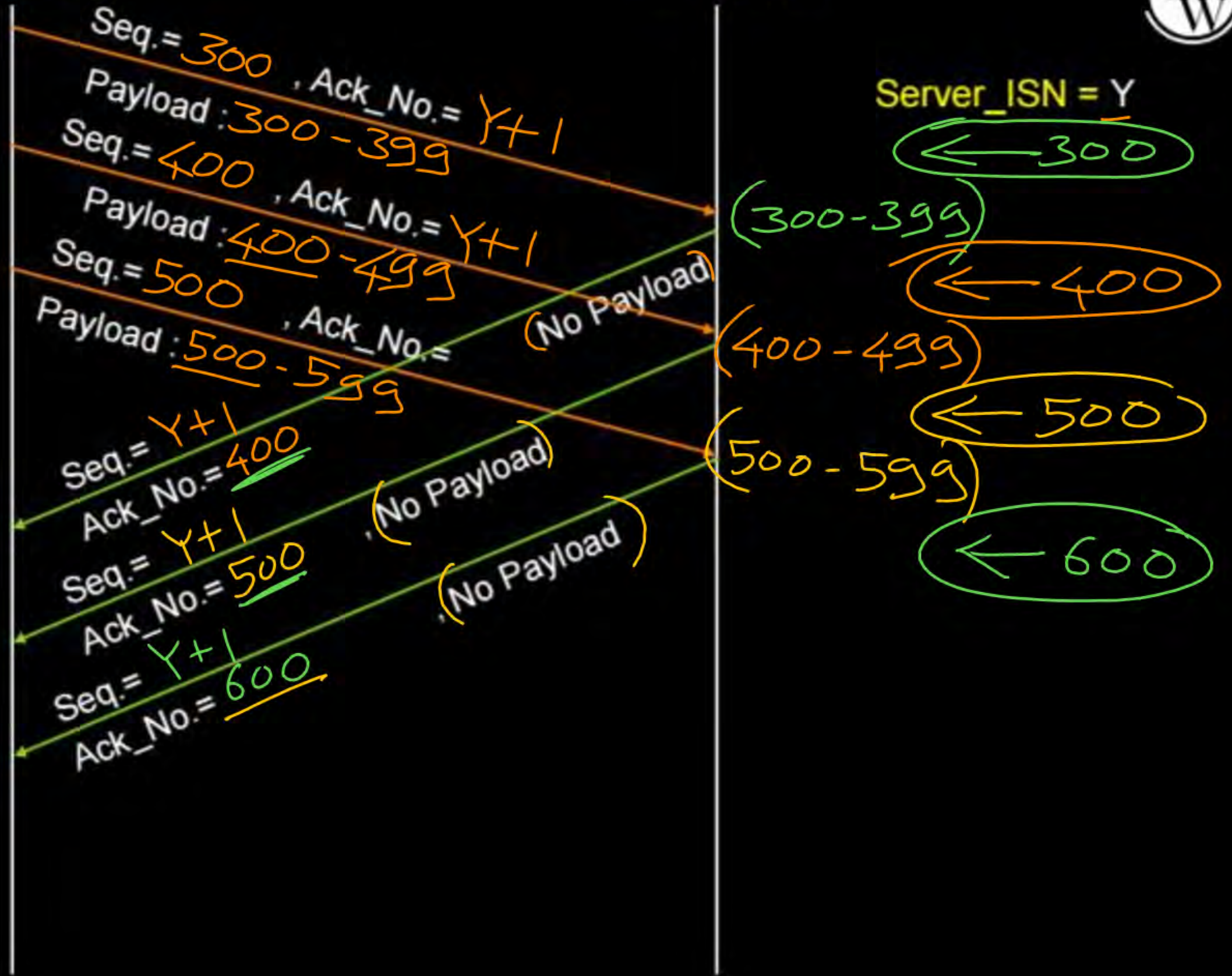
Topic : TCP

Client_ISN = 299

MSS = 100 bytes

TCP Client

TCP Server





Topic : TCP

Till $A-1$



=> The TCP protocol receive a segment from process P_2 ,
contains acknowledgement number A for process P_1

$$\rightarrow \text{LastByteAcked} = [A - 1;]$$

$\rightarrow$ Remove all the bytes of P_1 'Send Buffer' from begin to LastByteAcked



Topic : TCP

Client_ISN = 299

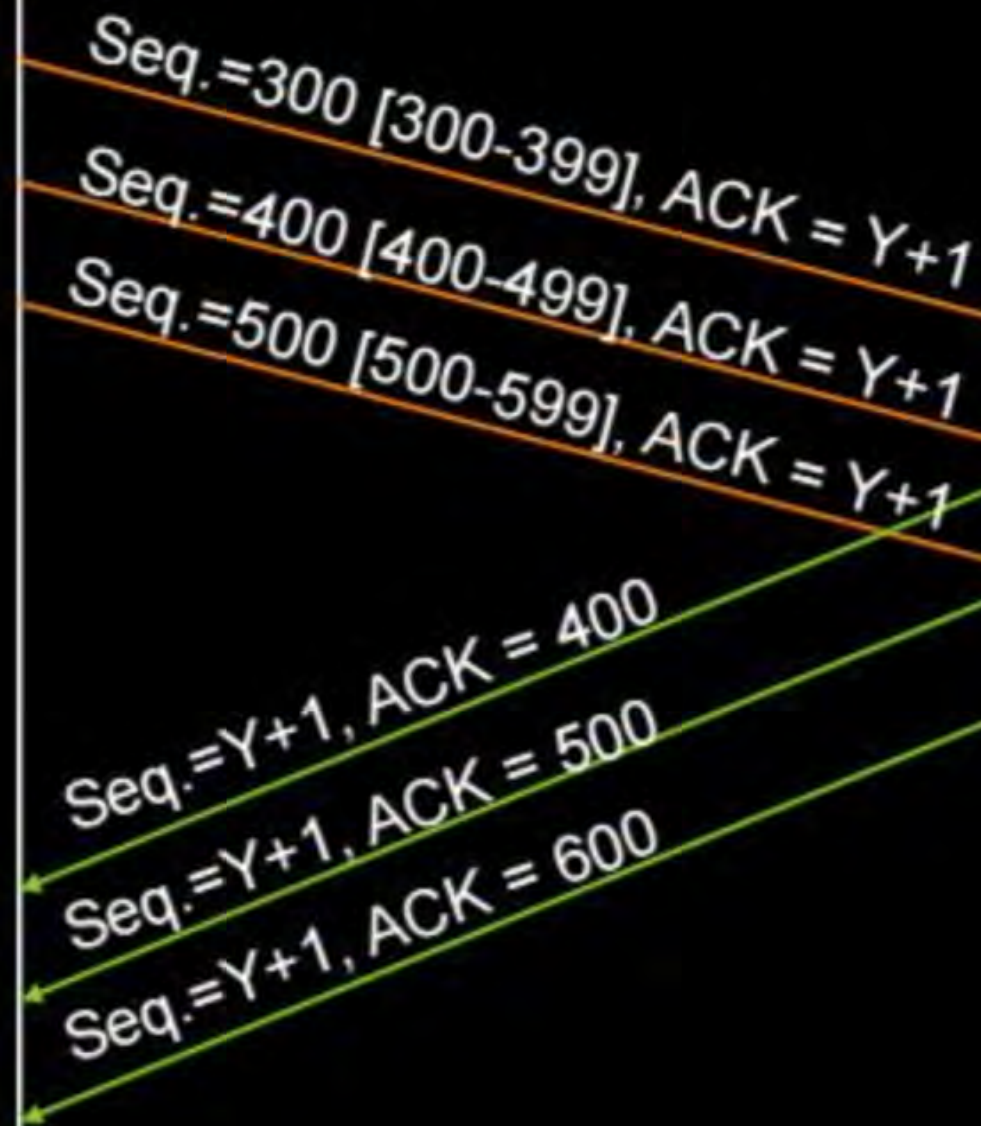
MSS = 100 bytes



TCP Client

TCP Server

Server_ISN = Y





2 mins Summary



Topic

TCP Data Transfer ✓



THANK - YOU

